

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which region of the small intestine is the main site for absorbing carbohydrates, proteins and lipids, and which takes up bile salts and vitamin B12?
2. Where are brush border enzymes found, and what do they finish digesting?
3. Which absorbed nutrients travel by lymph, and which go to the liver in the blood?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Structure and the three regions

- Longest part of the digestive tract: **pyloric sphincter → ileocecal junction**; ~3–4 m in the living person (~6.5 m cadaver), average diameter 2.5 cm; **principal site of nutrient digestion and absorption**.
- Four layers: **mucosa** (villi + microvilli ↑ surface area), **submucosa** (Brunner's glands), **muscularis** (circular + longitudinal smooth muscle), **serosa**. Folds, villi and microvilli give an extensive absorptive surface; **capillaries and lacteals** in the villi carry nutrients away.

Region	Key features
Duodenum	First and shortest (~25 cm); curves around the head of the pancreas; receives chyme plus secretions from liver, gallbladder and pancreas. Brunner's glands (submucosa) make alkaline mucus : neutralises acidic chyme, gives optimal pH for pancreatic enzymes, protects the mucosa. Endocrine cells secrete secretin + CCK
Jejunum	Middle (~2/5); specialised for absorption ; rich blood supply; prominent circular folds (plicae circulares) ; abundant villi. Main site for absorbing carbohydrates, proteins, lipids
Ileum	Longest (~3/5); fewer folds and villi than the jejunum but still absorbs significantly; uptake of bile salts and vitamin B12 ; ends at the ileocecal valve . Peyer's patches (B + T lymphocytes) give immune surveillance

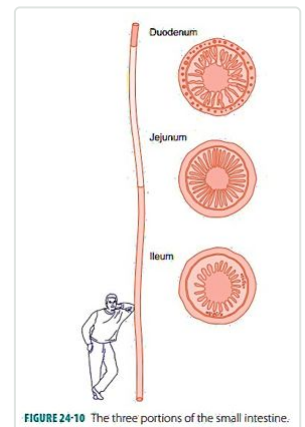
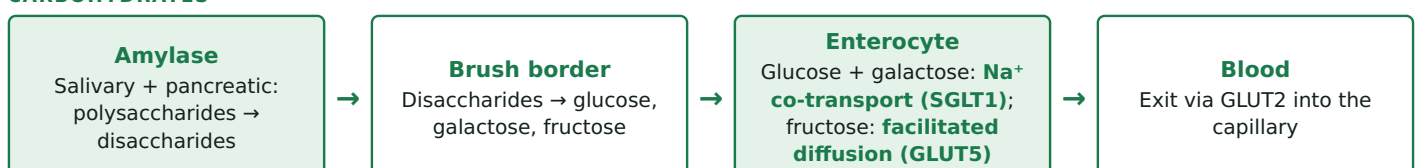


FIGURE 24-10 The three portions of the small intestine.

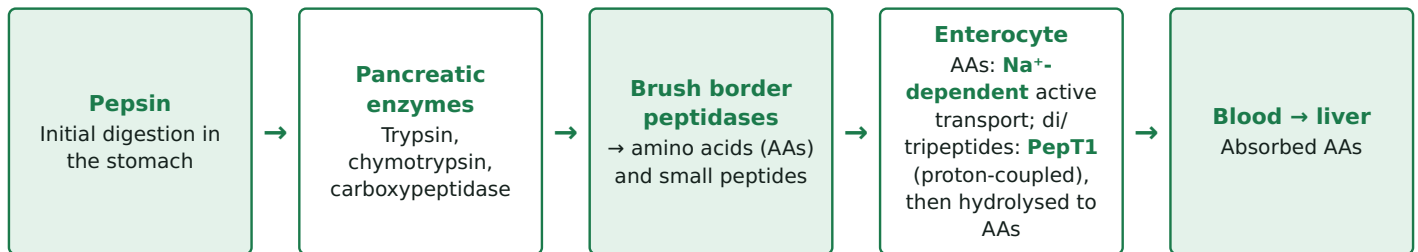
Duodenum, jejunum, ileum.

2 Digestive juice, brush border and motility

Feature	Detail
Intestinal juice	Watery mixture of mucus and enzymes ; the medium for digestion; enzymes complete carbohydrate and protein digestion; mucus lubricates the walls
Brush border enzymes	Lactase, maltase, sucrase, peptidases — bound to the microvillus membrane , not secreted into the lumen; give final hydrolysis (carbohydrates → glucose; peptides → amino acids)
Segmentation	Local contractions that mix chyme with digestive secretions and bring food into contact with the mucosa
Peristalsis	Waves of contraction that move chyme forward
Neural control	Enteric nervous system regulates motility and secretion; parasympathetic ↑ ↑ peristalsis + secretion; sympathetic ↓ ↓ motility + blood flow

3 Digestion and absorption chains**CARBOHYDRATES**

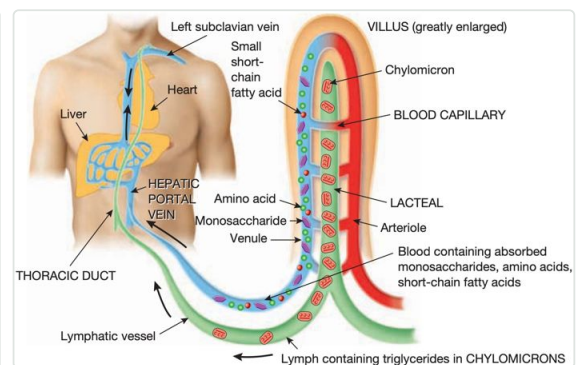
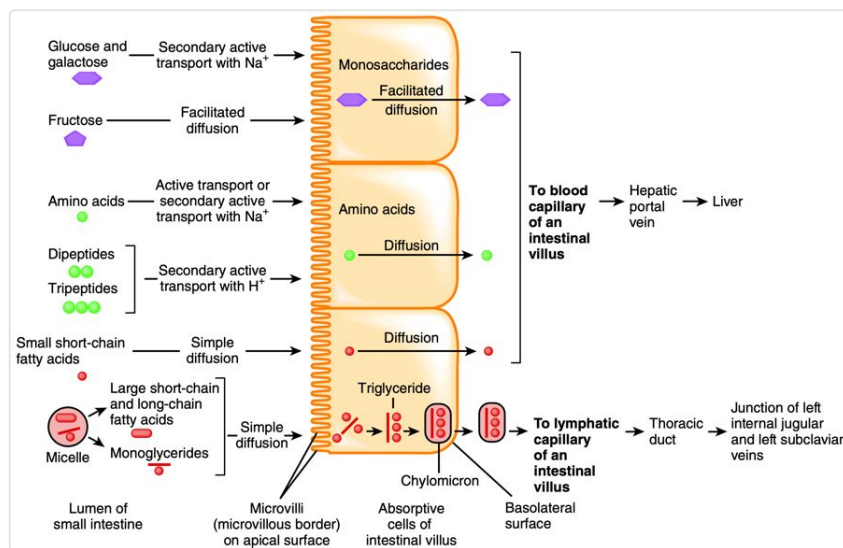
PROTEINS



LIPIDS



VISUAL ANCHOR · ACROSS THE ENTEROCYTE (LEFT) AND INTO BLOOD VS LYMPH (RIGHT)



Route	What travels this way
Blood (portal → liver)	Monosaccharides, amino acids, short-chain fatty acids
Lymph (lacteals → thoracic duct)	Triglycerides in chylomicrons

4 Vitamins, minerals, water and electrolytes

Item	Absorption
Fat-soluble A, D, E, K	With dietary lipids via micelles
Water-soluble vitamins	Diffusion or carrier-mediated mechanisms
Vitamin B12	Needs intrinsic factor ; absorbed in the ileum
Calcium	Duodenum ; vitamin D-dependent
Iron	Mainly duodenum ; enhanced by gastric acid
Mg, phosphate, zinc	Throughout the small intestine
Water + electrolytes	Large volumes daily. Na⁺ : coupled with nutrient absorption and via Na ⁺ channels; Cl⁻ : passive or exchanged with HCO ₃ ⁻ ; K⁺ : mainly diffusion

The small intestine is also an immune barrier (Peyer's patches, intraepithelial lymphocytes). Overall it **absorbs ~90%** of nutrients and water, completes digestion of carbohydrates, proteins and lipids, and begins and completes digestion of nucleic acids.

P Must-remember pearls

- **Duodenum** = shortest, Brunner's glands (alkaline mucus), secretin + CCK; **jejunum** = main absorptive site; **ileum** = bile salts, vitamin B12, Peyer's patches.
- **Brush border enzymes** (lactase, maltase, sucrase, peptidases) are **membrane-bound** and give final hydrolysis; only **monosaccharides** are absorbed.
- **Glucose + galactose** use **Na⁺-dependent co-transport**; **fructose** uses facilitated diffusion. **AAs** use Na⁺-dependent transport; **di/tripeptides** use **PepT1**.
- **Lipids**: bile salt micelles → lipase → diffuse in → re-esterify → **chylomicrons** → **lacteals (lymph)**. Sugars and amino acids go by **blood to the liver**.
- **Segmentation mixes; peristalsis propels**. Parasympathetic ↑ motility and secretion, sympathetic ↓. The small intestine absorbs **~90%** of nutrients and water.

Q Test yourself

1. A patient has damage to the glands in the duodenal submucosa. Which function is most directly lost?

- A. Secretion of secretin and cholecystokinin by endocrine cells
- B. Uptake of bile salts and vitamin B12
- C. Secretion of alkaline mucus that neutralises acidic chyme and protects the mucosa
- D. Immune surveillance against ingested pathogens
- E. Final hydrolysis of disaccharides at the microvilli

2. A student describes the brush border enzymes. Which statement is correct?

- A. They are bound to the microvillus membrane and give final hydrolysis of carbohydrates and peptides
- B. They are secreted by the stomach and act mainly on polysaccharides
- C. They are bound to the microvillus membrane and only digest lipids
- D. They are secreted into the lumen by the pancreas and act there
- E. They are stored in lacteals and released after a meal

3. Free amino acids reach the blood only after crossing the enterocyte. How are they taken up across the apical membrane?

- A. By facilitated diffusion through GLUT5
- B. By Na⁺-dependent active transport systems
- C. By the sodium-glucose cotransporter
- D. By simple diffusion through the lipid bilayer
- E. Inside micelles carried by bile salts

4. Which statement correctly compares the ileum with the jejunum?

- A. The ileum is identical in structure but lacks any absorptive role
- B. The ileum has more folds and villi and is the main site of absorption
- C. The ileum is the shortest region and receives bile and pancreatic secretions
- D. The ileum is longer and specialised only for immune function
- E. The ileum is longer (~3/5) with fewer folds and villi, yet still absorbs significantly

5. A student asks why the small intestine absorbs so efficiently. Which set of structures together provide the extensive absorptive surface?

- A. Villi, lacteals and the serosa
- B. Brunner's glands, villi and the submucosa
- C. Circular folds, Peyer's patches and Brunner's glands
- D. Circular folds, villi and microvilli
- E. Microvilli, the muscularis and the ileocecal valve

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — C** — Brunner's glands (duodenal submucosa) secrete alkaline mucus: neutralise chyme, optimal pH for pancreatic enzymes, protect mucosa.
- 2 — A** — Lactase, maltase, sucrase and peptidases are membrane-bound (not secreted) and complete carbohydrate and peptide digestion.
- 3 — B** — AAs are absorbed by Na⁺-dependent active transport; di- and tripeptides use PepT1 and are hydrolysed inside the cell.
- 4 — E** — Ileum ~3/5 of the length with fewer folds and villi than the jejunum but still contributes significantly to absorption.
- 5 — D** — Folds, villi and microvilli together form an extensive absorptive surface area; capillaries and lacteals carry nutrients away.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam